

Winning baseball games by solving statistics puzzles

Scott Powers

INFORMS Houston
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RICE UNIVERSITY
Sport Analytics

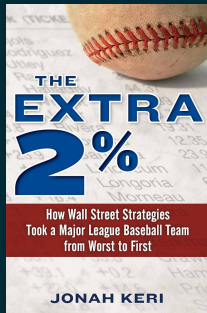
What is Sport Analytics?



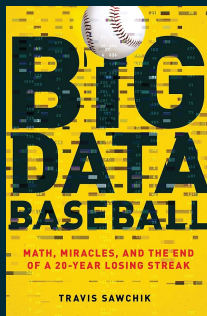
Books about Baseball Teams Using Analytics to Win



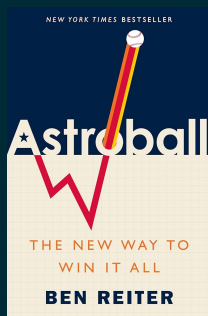
2002
Oakland
Athletics



2008
Tampa Bay
Rays



2013
Pittsburgh
Pirates



2017
Houston
Astros

MLB R&D Staff Sizes (according to Reddit)

Team	Analysts	Team	Analysts
American League		National League	
Yankees	20	Dodgers	20
Astros	15	Braves	15
Rays	15	Brewers	11
Angels	10	Reds	11
Tigers	9	Phillies	10
Rangers	8	Nationals	8
Mariners	7	Pirates	8
Royals	7	Padres	7
Twins	7	Cardinals	6
Blue Jays	6	Cubs	6
Red Sox	6	Giants	6
Indians	5	Marlins	6
Orioles	5	Diamondbacks	5
Athletics	3	Rockies	4
White Sox	2	Mets	3

2018

MLB Teams Ranked by Approximate R&D Size			
Includes Data Scientists/Analysts, Data/Software Engineers, Biomechanists, etc.			
TEAM	SIZE	TEAM	SIZE
	42		20
	41		20
	36		20
	35		20
	34		19
	34		19
	29		18
	28		17
	27		16
	26		16
	25		16
	24		15
	24		12
	23		12
	20		12
	20		10

Teams With No Clear Info: A's, Angels, Cubs, D-Backs

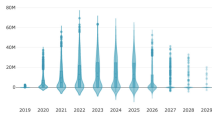
Amrit Vignesh | @avsportsanalyst

2025

What are they all doing?

Player Evaluation

Executives use modeling results to decide which players to target in trades and in free agency.



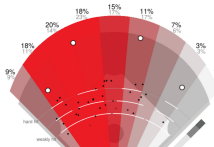
Player Development

Players use data-driven insights to tweak their game and achieve greater levels of performance.



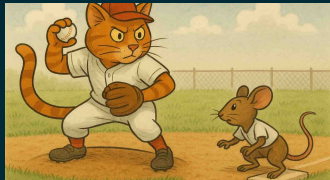
In-Game Strategy

Coaches use analytics tools to position fielders and decide when to make substitutions.

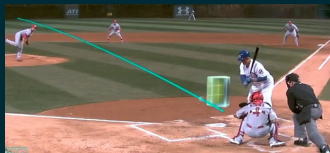


Three Puzzles

I: Runner-pitcher strategy under pickoff limits



II: Pitcher evaluation from pitch tracking



III: Inference from swing-level bat tracking

HOW LONG IS YOUR SWING?

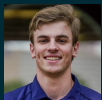
A SWING'S LENGTH IS CAPTURED FROM THE START OF A SWING UNTIL IMPACT POINT.

A SWING CAN BE AS SHORT AS 4 FEET OR AS LONG AS NEARLY 10 FEET.

A white line drawing of a baseball player in mid-swing. A white arrow starts at the beginning of the swing and points to the end of the bat at the moment of impact with the ball.

STATCAST powered by Google Cloud

Part I: Runner-pitcher strategy under pickoff limits



Jacob Hahn, Rice University → Chicago Cubs



Sivaramakrishnan Ramani, Rice University

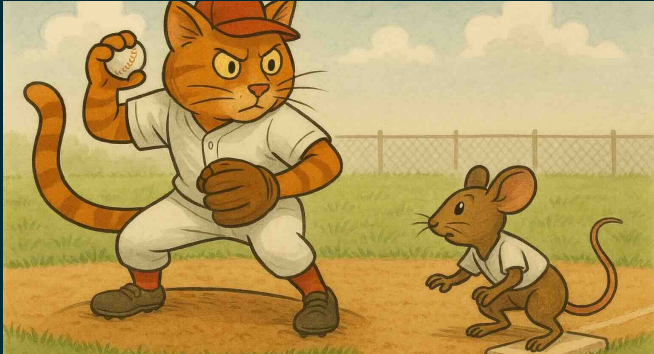


Andrew Schaefer, Rice University

Working paper:

Hahn J, Powers S, Ramani S, Schaefer A “The Two-Foot Rule: A game theoretic analysis of the pickoff limit in Major League Baseball”

Background



- In 2023, MLB limited the pitcher to 3 pickoff attempts
- After 3 failed pickoffs, the runner advances automatically

The Puzzle

If you are the runner, how much should you extend your lead after each pickoff attempt?

Data: pitch-by-pitch lead distance (player tracking) from MLB*

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Our Approach

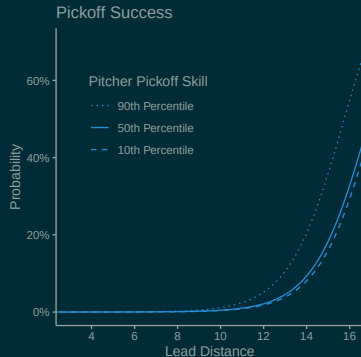
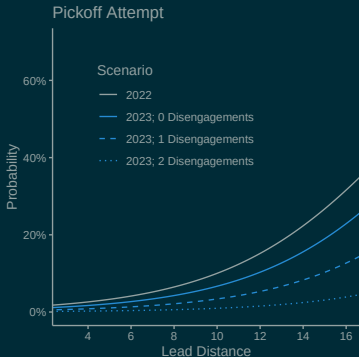
Step 1: Estimate the effect of lead distance

- Linear mixed-effects models (LMMs) for probabilities of pickoff attempt, pickoff success, stolen base success

Step 2: Solve for the optimal lead distance

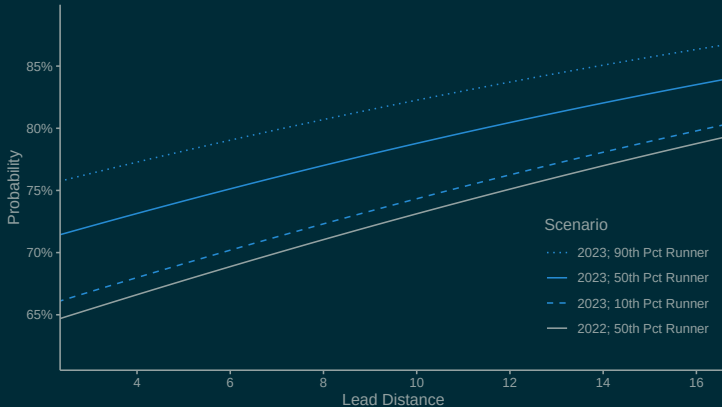
- Single-player MDP, or two-player stochastic game

Pickoff Attempt and Success Probability



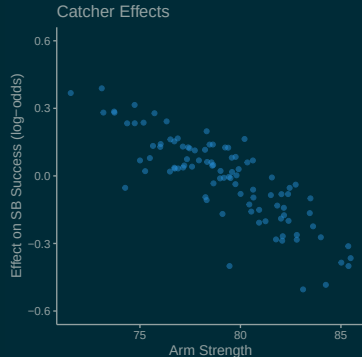
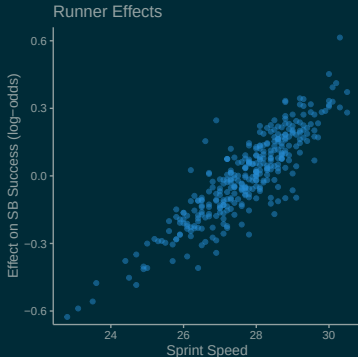
- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Stolen Base Success Probability



- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Runner and Catcher Effects



- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Solving for Optimal Lead Distance

Two options:

1. Single-player Markov decision process (MDP)
 - Runner chooses lead distance
 - Pitcher follows known stochastic response (no agency)
2. Two-player stochastic game
 - Runner chooses lead distance
 - Pitcher chooses whether to attempt pickoff

Optimal Lead Distance (single-player MDP)

Count	Disengagements		
	1	2	3
0-0	10.0	11.5	14.1
1-0	10.2	11.7	14.2
2-0	10.2	11.7	14.2
3-0	10.3	11.8	14.2
0-1	9.8	11.4	14.1
1-1	10.0	11.6	14.2
2-1	10.3	11.8	14.3
3-1	10.4	11.9	14.3
0-2	9.5	11.1	14.1
1-2	9.9	11.4	14.3
2-2	10.0	11.5	14.2
3-2	10.4	11.9	14.4

- **The Two-Foot Rule:**
Increase lead by two feet after each disengagement
- Value of implementation:
~17 runs / team / season
(~\$17,000,000)

Optimal Lead Distance (two-player stochastic game)

Count	Disengagements		
	1	2	3
0-0	11.2	11.8	16.2
1-0	11.2	12.0	16.1
2-0	11.0	12.2	15.9
3-0	10.4	11.6	15.3
0-1	10.8	11.3	16.2
1-1	10.9	11.1	16.1
2-1	11.0	12.1	15.9
3-1	10.5	11.6	15.7
0-2	10.2	10.8	16.2
1-2	10.5	10.4	16.3
2-2	10.4	11.4	15.9
3-2	10.3	11.1	15.6

- Equilibrium entails more aggressive behavior from both runner and pitcher

Part II: Pitcher evaluation from pitch tracking

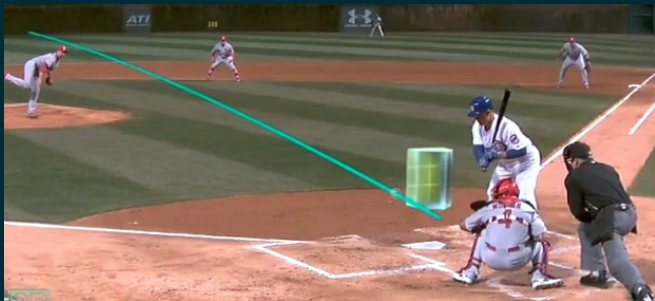


Vicente Iglesias, Susquehanna International Group

Working paper:

Powers S, Iglesias V “Pitch trajectory density estimation for predicting future outcomes”

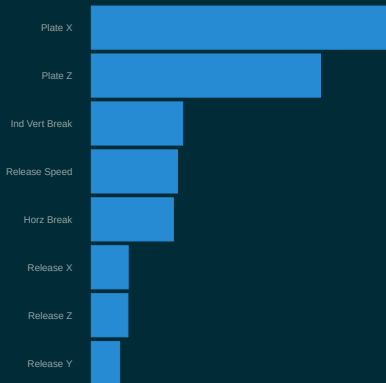
Background



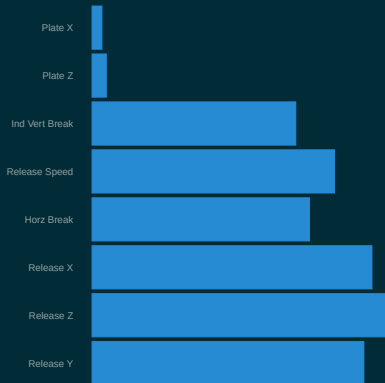
- Full pitch tracking data have been available for two decades
- Trajectory estimated as 3D quadratic function of time ($\in \mathbb{R}^9$)
- Standard practice: Evaluate pitchers based on trajectories

The Puzzle

Variable Importance¹



Variable Reliability²



¹ fractional contribution of each feature's splits to gradient boosting pitch model

² (between-pitcher variance) / (total variance); varies by pitch type (here: RHB FB)

Why Supervised Learning Isn't Enough

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
$\dots$		
x_n	$\rightarrow$	y_n

$$x^* \rightarrow \hat{y} = \hat{f}(x)$$

Our Problem

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
	$\dots$		
C	x_n	$\rightarrow$	y_n



A

$\hat{y}$

Why Supervised Learning Isn't Enough

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
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$$x^* \rightarrow \hat{y} = \hat{f}(x)$$

Our Approach

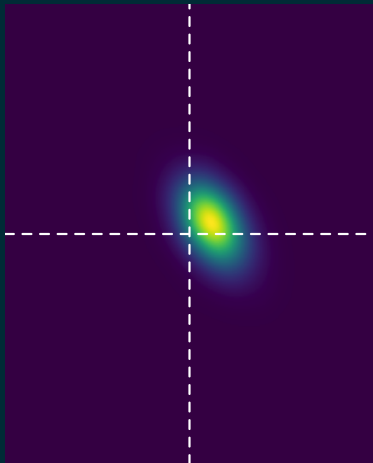
Pitcher	Pitch		Outcome
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	$\dots$		
C	x_n	$\rightarrow$	y_n



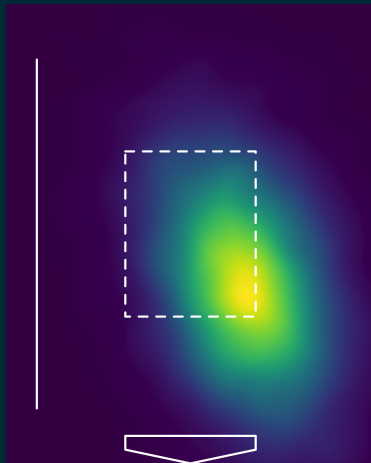
$$A \quad \hat{p}(x|A) \rightarrow \int \hat{p}(x) \hat{f}(x)$$

Dylan Cease's Slider vs RHB in 0-0 Counts

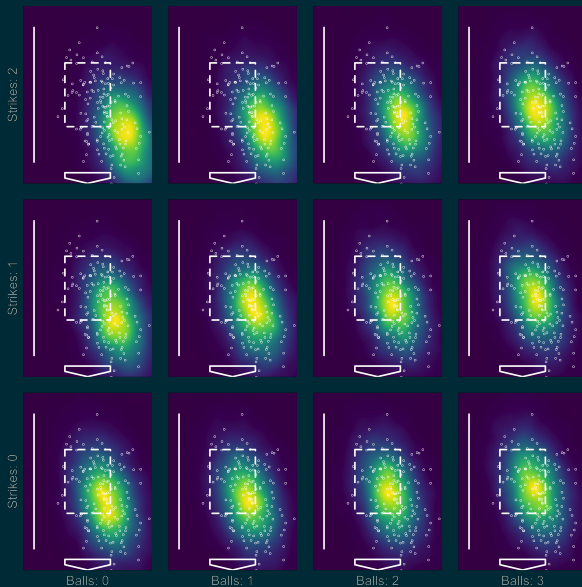
Predicted Break Chart



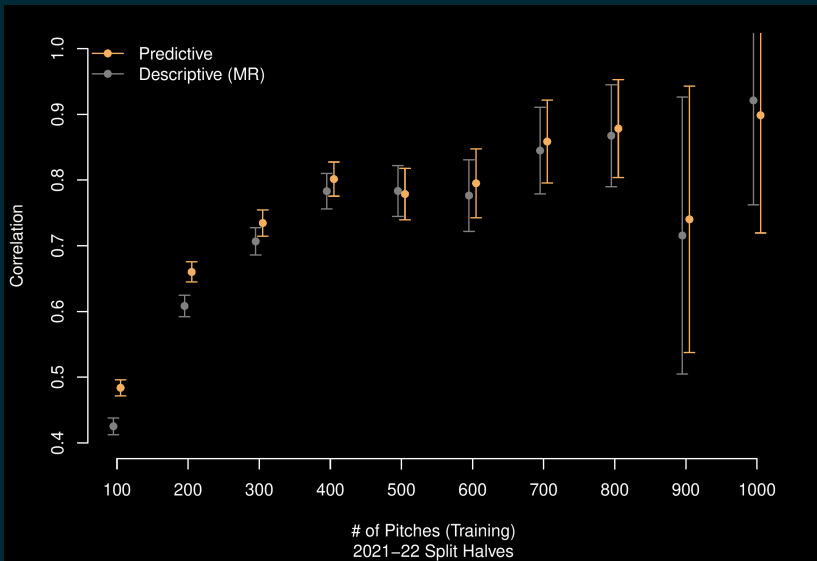
Predicted Plate Location



Dylan Cease's Slider vs RHB in All Counts



Predictive Performance



Part III: Inference from swing-level bat tracking



Ron Yurko, Carnegie Mellon University

Working paper:

Powers S, Yurko R "Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics" arXiv:2507.01238

Background

HOW LONG IS YOUR SWING?

A SWING'S LENGTH IS CAPTURED
FROM THE START OF A SWING UNTIL
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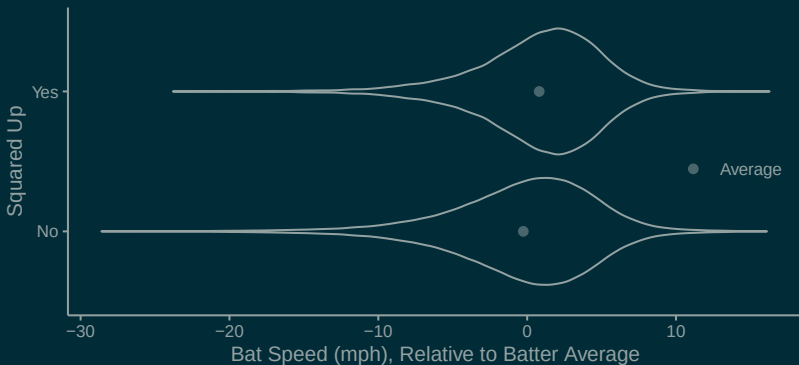


savant

- In May 2024, MLB released swing length and bat speed data
- Both are measured at contact (or at point nearest to contact)

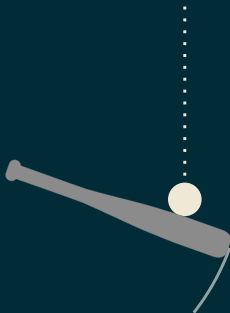
The Puzzle

"Squared Up" = good contact (exit velocity > 80% of theoretical max)

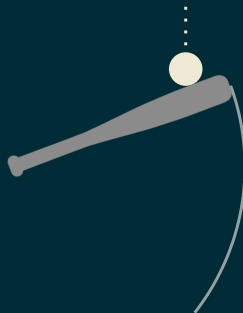


- How should batters modulate swing length and bat speed?

Challenge #1: Measurement



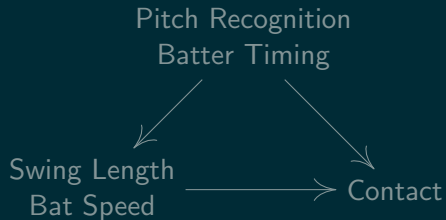
LATE Swing



EARLY Swing

- For **identical** swings, timing determines point of measurement
- Swing length and bat speed are **outcomes**, not purely process

Challenge #2: Confounding



Our Approach

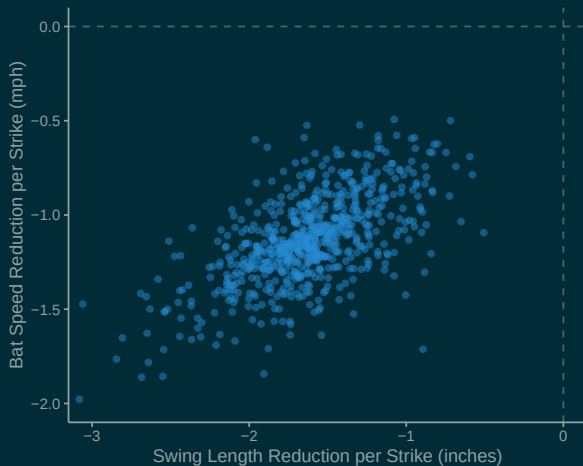
Step 1: Estimate batter intention

- Bayesian hierarchical model (skew-normal likelihood)
- Condition on successful contact against fastballs!

Step 2: Estimate effect of batter intention

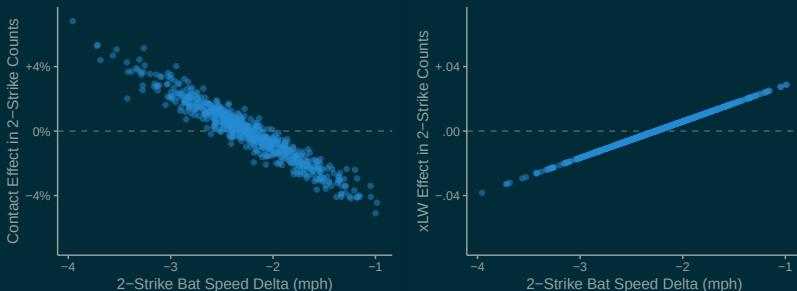
- Instrumental variable regression (instrument: count)

Batter $\times$ Count Interaction



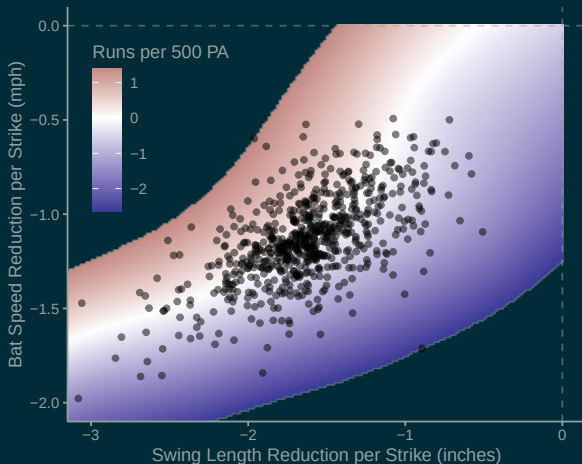
- Effects: batter, count, location, batter $\times$ (count, location)
- We observe heterogeneity in how batters adjust to count

Causal Effect Estimates



- Batters who slow down more with two strikes exhibit **more contact** (relative to expectation) with two strikes
- Batters who slow down more with two strikes exhibit **less power** (relative to expectation) with two strikes
- Is the tradeoff worthwhile?

Contact-Power Tradeoff



- Batters can reduce strikeouts by slowing their bat speeds, but the tradeoff in power makes it a wash

